

Expt 3	Performance Analysis of IIR Filters
Date	3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

CODE :-

```

clc;

clear;

close all;

% Specifications

Fs = 2000;    % Sampling Frequency

Rp = 1;       % Passband Ripple (dB)

Rs = 40;      % Stopband Attenuation (dB)

Wp = [300 600]/(Fs/2);  % Passband Frequencies

Ws = [200 700]/(Fs/2);  % Stopband Frequencies

%% Butterworth Band-Pass Filter

[n1,W1] = buttord(Wp,Ws,Rp,Rs);

[b1,a1] = butter(n1,W1,'bandpass');

%% Chebyshev Type-I Band-Pass Filter

[n2,W2] = cheb1ord(Wp,Ws,Rp,Rs);

[b2,a2] = cheby1(n2,Rp,W2,'bandpass');

%% Display Filter Orders

disp(['Butterworth Filter Order = ', num2str(n1)]);

disp(['Chebyshev Filter Order = ', num2str(n2)]);

%% Frequency Response

figure;

freqz(b1,a1);

title('Butterworth Band-Pass Filter');

figure;

freqz(b2,a2);

```

```

title('Chebyshev Type-I Band-Pass Filter');

%%% Group Delay

figure;

grpdelay(b1,a1);

hold on;

grpdelay(b2,a2);

legend('Butterworth','Chebyshev-I');

title('Group Delay Comparison');

grid on;

%%% Pole-Zero Plot

figure;

subplot(1,2,1);

zplane(b1,a1);

title('Butterworth Pole-Zero Plot');

subplot(1,2,2);

zplane(b2,a2);

title('Chebyshev Type-I Pole-Zero Plot');

%%% Impulse Response

figure;

subplot(2,1,1);

impz(b1,a1);

title('Butterworth Impulse Response');

subplot(2,1,2);

impz(b2,a2);

title('Chebyshev Type-I Impulse Response');

%%% Pole Magnitudes

figure;

subplot(1,2,1);

stem(abs(roots(a1)),'filled');

```

```

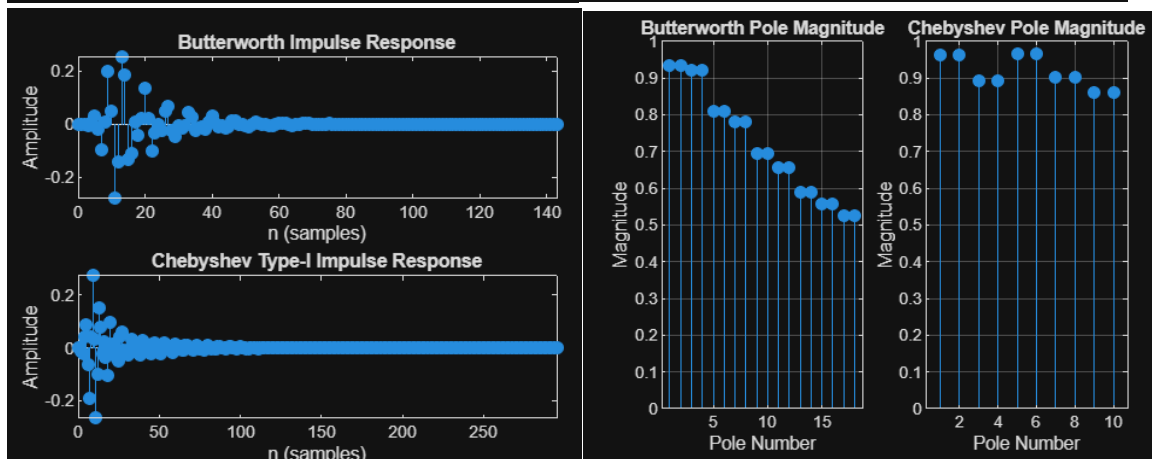
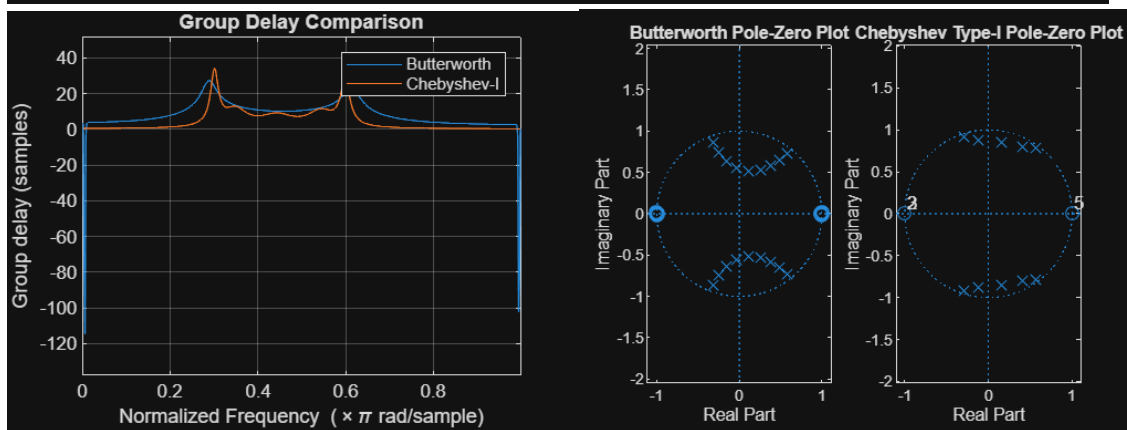
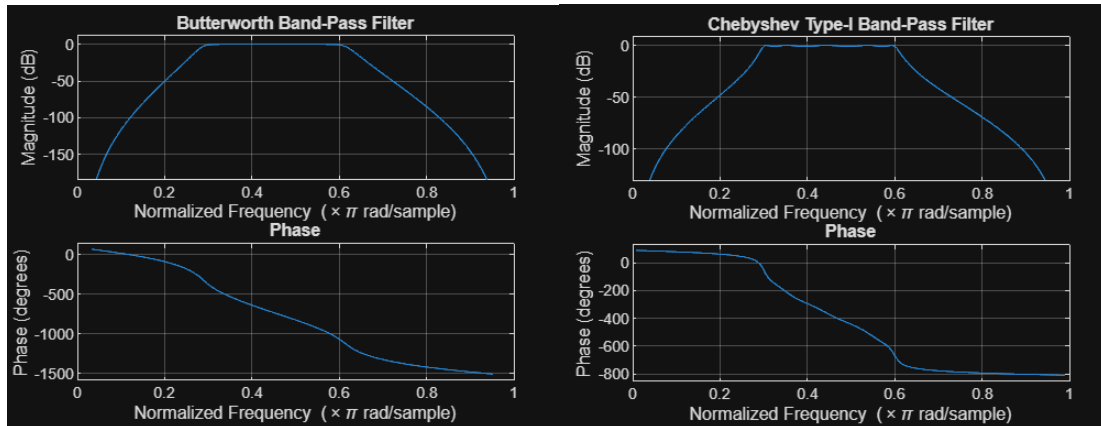
title('Butterworth Pole Magnitude');
xlabel('Pole Number');
ylabel('Magnitude');
grid on;
subplot(1,2,2);
stem(abs(roots(a2)),'filled');
title('Chebyshev Pole Magnitude');
xlabel('Pole Number');
ylabel('Magnitude');
grid on;
%% Stability Check
if all(abs(roots(a1)) < 1)
    disp('Butterworth Filter is STABLE');
else
    disp('Butterworth Filter is UNSTABLE');
end
if all(abs(roots(a2)) < 1)
    disp('Chebyshev Filter is STABLE');
else
    disp('Chebyshev Filter is UNSTABLE');
end

```

OUTPUT :-

Butterworth Filter Order = 9

Chebyshev Filter Order = 5



Butterworth Filter is STABLE

Chebyshev Filter is STABLE